THRESHOLD BASED ANOMALY DETECTION FOR OVER-THE-AIR THREAT MITIGATION

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Declaration page of the candidate & supervisor

I declare that this is my own work, and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute my dissertation in whole or part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as article or books).

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Abstract

The rapid expansion of Internet of Things (IoT) ecosystems has led to widespread deployment of interconnected devices across critical domains such as healthcare, industrial automation, smart cities, and consumer applications. As the scale and complexity of these systems increase, ensuring secure, reliable, and efficient firmware update mechanisms has become a major challenge. Over-the-Air (OTA) update systems are widely adopted for remote firmware management; however, existing solutions primarily focus on secure transmission and integrity verification of update packages, while largely ignoring the operational state of devices prior to update execution. This limitation exposes IoT systems to significant risks, as firmware updates applied to unstable or compromised devices may result in system failures, corruption, or security breaches.

This research proposes a Secure Trust OTA framework that integrates a threshold-based anomaly detection mechanism into the firmware update lifecycle to ensure that updates are executed only when devices are operating under stable and trusted conditions. The proposed system continuously monitors key device parameters such as CPU usage, memory consumption, battery level, and network performance through lightweight telemetry agents. These parameters are evaluated against predefined and adaptive threshold values to detect abnormal behavior in real time. A gating mechanism is introduced to block firmware updates for devices exhibiting anomalous conditions, thereby enhancing system reliability and preventing unsafe update execution.

The methodology follows a modular system design approach comprising telemetry collection, secure communication, anomaly detection, OTA update management, and alert generation. The system is implemented using lightweight technologies suitable for resource-constrained IoT environments, ensuring scalability and efficiency. Comprehensive testing is conducted under various normal and abnormal operational scenarios to evaluate system performance, detection accuracy, and response latency.

The results demonstrate that the proposed framework effectively identifies anomalous device behavior with high accuracy while maintaining low computational overhead and minimal latency. The integration of anomaly detection into the OTA process significantly reduces the risk of update failures and enhances overall system stability. Furthermore, the system aligns with Zero Trust security principles by enforcing continuous validation of device trustworthiness before executing critical operations.

In conclusion, this research presents a practical and scalable approach to improving the security and reliability of OTA firmware updates in IoT environments. The proposed

framework bridges the gap between anomaly detection and update management, offering a proactive solution that enhances operational safety and contributes to the advancement of secure IoT system design.

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I would like to express my sincere gratitude to all those who have supported and guided me throughout the completion of this final year research project. This work would not have been possible without their valuable assistance, encouragement, and continuous motivation.

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Table of Contents

List of Figures

List of Tables

Introduction

The rapid evolution of digital technologies has led to the emergence of the Internet of Things (IoT) as a transformative paradigm that connects physical devices to the digital world. IoT systems are increasingly being deployed across a wide range of sectors, including healthcare, industrial automation, transportation, agriculture, and smart city infrastructure. These systems rely on interconnected devices that continuously collect, process, and exchange data to enable intelligent decision-making and automation. As the scale and complexity of IoT deployments continue to grow, ensuring the reliability, security, and maintainability of these devices has become a critical challenge for researchers and industry practitioners.

One of the key requirements in managing IoT systems is the ability to update device firmware efficiently and securely. Firmware updates are essential for fixing software bugs, addressing newly discovered vulnerabilities, improving performance, and introducing new functionalities. Over-the-Air (OTA) update mechanisms have emerged as a widely adopted solution for remotely managing firmware updates without requiring physical access to devices. OTA systems provide scalability and convenience, particularly in large-scale deployments where devices are geographically distributed. However, despite these advantages, existing OTA frameworks primarily focus on ensuring secure delivery of firmware packages through encryption and integrity verification, while often neglecting the operational state of devices receiving these updates.

Applying firmware updates to devices without considering their current condition introduces significant risks. IoT devices may operate under abnormal conditions due to factors such as hardware failures, resource exhaustion, network instability, or cyberattacks. In such situations, executing firmware updates can lead to incomplete installations, system crashes, data corruption, or permanent device failure. These risks highlight a critical limitation in current OTA systems, where the lack of device state validation undermines the reliability and safety of the update process.

To address this issue, anomaly detection has been widely studied as a method for identifying abnormal behavior in systems. Traditional approaches often rely on machine learning and deep learning techniques to analyze large volumes of data and detect complex patterns. While these methods can achieve high accuracy, they are computationally expensive and require significant resources for training and deployment. This makes them less suitable for IoT environments, where devices are typically resource-constrained. As a result, there is a growing need for lightweight and efficient anomaly detection techniques that can operate in real time without imposing high computational overhead.

This thesis proposes a Secure Trust OTA framework that integrates a threshold-based anomaly detection mechanism into the firmware update process. The core idea is to continuously monitor key operational parameters of IoT devices, such as CPU usage, memory utilization, battery levels, and network performance, and evaluate them against predefined and adaptive thresholds. By doing so, the system is able to determine whether a device is operating under normal or abnormal conditions. A gating mechanism is then applied to ensure that firmware updates are only executed when devices are in a stable and trusted state. This approach introduces a proactive layer of validation that enhances both reliability and security.

The research adopts a structured methodology that includes system design, implementation, testing, and evaluation. A modular architecture is developed to integrate telemetry collection, secure communication, anomaly detection, and OTA update management into a unified framework. The system is implemented using lightweight technologies to ensure scalability and efficiency, making it suitable for large-scale IoT deployments. Extensive testing is conducted under various scenarios to evaluate system performance, detection accuracy, and operational reliability.

The contributions of this thesis lie in bridging the gap between anomaly detection and OTA update mechanisms by providing a practical and efficient solution that enhances system trustworthiness. The proposed framework not only improves the reliability of firmware updates but also aligns with modern cybersecurity principles such as Zero Trust, which emphasizes continuous validation rather than implicit trust. By ensuring that updates are applied only under safe conditions, the system reduces the risk of failures and strengthens the overall security posture of IoT environments.

Background & Literature Survey

The rapid growth of the Internet of Things (IoT) has revolutionized the way digital systems interact with the physical world by enabling seamless connectivity between billions of devices. These devices, ranging from simple sensors to complex embedded systems, are increasingly being deployed in critical sectors such as healthcare, industrial automation, transportation, agriculture, and smart city infrastructure. IoT systems are designed to operate continuously, often in real-time environments, where reliability, availability, and security are essential for maintaining operational integrity. As the number of connected devices continues to grow exponentially, managing and maintaining these systems has become a significant challenge.

One of the fundamental requirements in IoT ecosystems is the ability to update device firmware efficiently and securely. Firmware updates are necessary to fix software bugs, patch newly discovered vulnerabilities, enhance device performance, and introduce new features. Traditionally, firmware updates required physical access to devices, which is impractical in large-scale or geographically distributed deployments. To overcome this limitation, Over-the-Air (OTA) update mechanisms have been widely adopted as a scalable solution that allows remote firmware updates via network connectivity. OTA updates enable centralized control and management of devices, significantly reducing maintenance costs and improving operational efficiency.

Despite their advantages, OTA update mechanisms introduce several security and reliability concerns. Existing OTA frameworks primarily focus on ensuring secure transmission of firmware packages using cryptographic techniques such as Transport Layer Security (TLS), encryption, and digital signatures. These mechanisms ensure that firmware updates are authentic and have not been tampered with during transmission. However, they do not consider the operational state of the device receiving the update. In real-world scenarios, IoT devices may operate under abnormal conditions due to factors such as malware infections, hardware malfunctions, resource exhaustion, network instability, or misconfigurations. Applying updates to devices in such unstable states can lead to severe consequences, including incomplete updates, system crashes, firmware corruption, and even permanent device failure.

The literature highlights that firmware updates themselves are a major attack vector in IoT ecosystems. Attackers can exploit vulnerabilities in the update process to inject malicious firmware, perform downgrade attacks, or disrupt update operations. Several studies emphasize the importance of securing OTA mechanisms through authentication, encryption, and integrity verification. However, these approaches primarily address the security of the update package rather than the trustworthiness of the device receiving it. This creates a critical gap in ensuring end-to-end security in OTA processes.

In parallel, anomaly detection has emerged as a key research area in IoT security, focusing on identifying abnormal behavior that may indicate system faults or cyber threats. Machine learning and deep learning techniques have been widely used for anomaly detection due to their ability to identify complex patterns in large datasets. Approaches such as clustering, neural networks, and autoencoders have demonstrated high accuracy in detecting anomalies in IoT systems. However, these methods require significant computational resources, large volumes of training data, and continuous model updates, making them unsuitable for deployment in resource-constrained IoT environments.

As an alternative, threshold-based anomaly detection methods have gained attention due to their simplicity, efficiency, and suitability for real-time monitoring. These methods rely on predefined limits for key operational parameters such as CPU usage, memory utilization, battery level, and network activity. When a parameter exceeds or falls below its threshold, it is considered anomalous. Threshold-based approaches are highly interpretable, require minimal computational overhead, and can be easily implemented in embedded systems. Despite these advantages, their application has largely been limited to general monitoring and intrusion detection, with little focus on integrating them into OTA update workflows.

Another important concept in modern cybersecurity is the Zero Trust model, which eliminates implicit trust and requires continuous verification of all entities within a system. While Zero Trust principles have been widely applied in network security and access control, their integration into OTA update processes remains limited. Most OTA systems implicitly trust devices once they are authenticated, without continuously validating their operational state before performing critical actions such as firmware updates.

Therefore, the existing literature indicates a clear need for a solution that combines secure OTA mechanisms with lightweight, real-time anomaly detection and continuous trust validation. This research proposes such a solution by integrating a threshold-based anomaly detection engine into the OTA update process, ensuring that updates are applied only to devices operating within safe and stable conditions.

Research Gap

Despite the extensive research conducted in the fields of IoT security, anomaly detection, and OTA update mechanisms, a significant gap remains in the integration of these domains. Current OTA frameworks, including widely used platforms such as Mender and Eclipse Hawkbit, are designed to ensure secure firmware delivery and provide features such as rollback recovery and device management. However, these systems operate under the assumption that devices are functioning normally before updates are applied. This assumption is unrealistic in practical environments, where devices may frequently experience anomalies due to various internal and external factors.

Existing anomaly detection solutions in IoT environments primarily focus on identifying malicious activities, system faults, or abnormal network behavior. While these solutions are effective in detecting anomalies, they are typically implemented as standalone monitoring systems and are not integrated into the OTA update decision-making process. As a result, even if anomalies are detected, they do not directly influence whether a device should receive a firmware update. This lack of integration creates a critical vulnerability, as updates may still be applied to compromised or unstable devices.

Another major gap lies in the reliance on machine learning-based approaches for anomaly detection. Although these methods provide high accuracy, they are computationally expensive and require significant resources for training and deployment. This makes them impractical for many IoT devices, which operate under strict resource constraints. Lightweight threshold-based methods offer a more suitable alternative, but their potential has not been fully explored in the context of OTA security.

Additionally, most existing systems use static threshold values that do not adapt to changing device behavior or environmental conditions. This can lead to false positives or missed anomalies, reducing the effectiveness of detection mechanisms. The lack of adaptive thresholding and centralized management further limits the scalability and flexibility of these systems.

Furthermore, the application of Zero Trust principles in OTA workflows is still underdeveloped. Current systems do not continuously verify device health before performing updates, resulting in implicit trust that can be exploited by attackers. This highlights the need for a more comprehensive approach that incorporates continuous validation, anomaly detection, and secure update mechanisms.

In summary, the research gap can be identified as the absence of a lightweight, adaptive, and integrated anomaly detection mechanism within OTA frameworks that validates device trustworthiness before updates, particularly in resource-constrained IoT environments.

Table 1: Research Gap Comparison

Area	Existing Approaches	Limitations Identified	Research Gap	Proposed Solution in This Study
OTA Update Mechanisms	Secure OTA frameworks use encryption, digital signatures, and secure protocols to deliver firmware updates.	Focus mainly on secure transmission and integrity of firmware, ignoring device operational state before updates.	Lack of validation of device health prior to update execution.	Introduce a trust-based OTA framework that evaluates device condition before allowing updates.
Firmware Update Reliability	Rollback and recovery mechanisms are used to handle update failures.	Reactive approach—failures are handled after they occur rather than being prevented.	No proactive mechanism to prevent updates on unstable devices.	Implement a gating mechanism to block updates when anomalies are detected.
Anomaly Detection in IoT	Machine learning and deep learning-based anomaly detection techniques.	High computational cost, requires training data, not suitable for resource-constrained IoT devices.	Lack of lightweight, real-time anomaly detection suitable for IoT environments.	Use threshold-based anomaly detection for efficient and real-time monitoring.

Threshold-Based Detection	Basic threshold systems used for monitoring system metrics.	Static thresholds lead to false positives/negatives and lack adaptability.	Absence of adaptive thresholding tailored to dynamic IoT behavior.	Implement adaptive thresholding using historical telemetry data.
Integration of Detection with OTA	Anomaly detection systems exist as separate monitoring tools.	Detection results are not integrated into OTA update decision-making.	No direct link between anomaly detection and firmware update control.	Integrate anomaly detection engine directly into OTA update workflow.
Scalability in IoT Systems	Centralized OTA systems manage large device fleets.	High load on central servers and potential latency issues.	Limited scalability and real-time responsiveness in large deployments.	Design lightweight and scalable architecture with efficient processing.
Real-Time Decision Making	Batch processing or delayed anomaly detection approaches.	Delayed response to anomalies may allow unsafe updates.	Lack of real-time anomaly detection and decision-making.	Enable real-time telemetry analysis and instant update gating decisions.

Research Problem

The increasing reliance on IoT devices in critical applications has amplified the importance of secure and reliable firmware update mechanisms. While OTA updates provide a convenient and efficient solution for managing device software, they introduce significant risks when applied to devices operating under abnormal conditions. Existing OTA frameworks focus primarily on securing the update package through encryption and integrity verification but fail to consider the operational state of the device receiving the update.

In real-world scenarios, IoT devices may exhibit anomalous behavior due to cyberattacks, hardware failures, resource exhaustion, or environmental factors. Applying updates under such conditions can lead to unpredictable outcomes, including system instability, failed updates, propagation of malicious code, and permanent device damage. These risks are further exacerbated by the lack of integration between anomaly detection systems and OTA update processes.

Although anomaly detection techniques exist, most are based on machine learning models that are not suitable for resource-constrained IoT devices. Lightweight threshold-based methods, while more practical, have not been effectively integrated into OTA frameworks to influence update decisions.

Therefore, the central research problem addressed in this study is the lack of a lightweight, efficient, and integrated mechanism to evaluate device trustworthiness before OTA updates, resulting in increased vulnerability to failures and security threats in IoT systems.

Research Objectives

The main objective of this research is to develop a secure, reliable, and trust-based OTA update framework for IoT devices that ensures firmware updates are applied only when devices are operating in a stable and trusted state. This objective is achieved by integrating a threshold-based anomaly detection mechanism into the OTA update lifecycle.

To accomplish this goal, several specific objectives are defined. First, the research aims to design and implement a telemetry collection system that continuously gathers operational data from IoT devices, including CPU usage, memory utilization, battery levels, network performance, and error rates. This data serves as the foundation for anomaly detection and decision-making.

Second, the study focuses on defining and managing threshold values that represent normal operating conditions for different device types. These thresholds are used to identify deviations in device behavior and detect anomalies in real time. The research also explores adaptive thresholding techniques to improve accuracy and reduce false positives.

Third, the research aims to develop a lightweight anomaly detection engine that processes telemetry data and determines whether a device is operating within acceptable limits. This engine is designed to be efficient and suitable for deployment in resource-constrained environments.

Another key objective is to integrate the anomaly detection mechanism into the OTA update process, enabling a gating mechanism that allows updates only for devices that meet predefined stability criteria. Additionally, the system is designed to generate alerts and provide monitoring capabilities to support real-time decision-making and system management.

Finally, the research aims to evaluate the effectiveness of the proposed framework through comprehensive testing and analysis, assessing its performance, accuracy, and impact on system reliability and security.

Methodology

The methodology adopted in this research focuses on designing and implementing a secure, resilient, and trust-aware Over-the-Air (OTA) firmware update framework that integrates a threshold-based anomaly detection engine to validate device health before updates are applied. The overall approach follows a system engineering methodology that combines architectural design, data-driven decision-making, and security principles such as Zero Trust to address the challenges identified in IoT firmware update mechanisms. The core objective of the methodology is to ensure that firmware updates are not only securely delivered but also safely executed under stable device conditions, thereby minimizing risks associated with updating compromised or unstable devices.

The proposed system operates as a continuous monitoring and validation pipeline where IoT devices generate telemetry data that reflects their operational state. This data is transmitted to a centralized OTA server through secure communication channels, where it is analyzed using predefined threshold rules. The methodology emphasizes a lightweight and deterministic approach rather than computationally intensive machine learning techniques, ensuring that the solution remains feasible for resource-constrained environments. The detection process is integrated directly into the OTA update workflow, creating a gating mechanism that allows updates only when devices satisfy predefined stability conditions.

The design phase of the methodology involves identifying critical system components, defining data flows, and establishing communication protocols between devices and the server. The implementation phase focuses on developing modules for telemetry collection, anomaly detection, update management, and alert generation. The evaluation phase includes testing the system under various scenarios to assess its performance, reliability, and effectiveness in detecting anomalies and preventing unsafe updates. This structured approach ensures that the proposed solution is both practical and scalable, capable of addressing real-world IoT deployment challenges.

Overall System Architecture

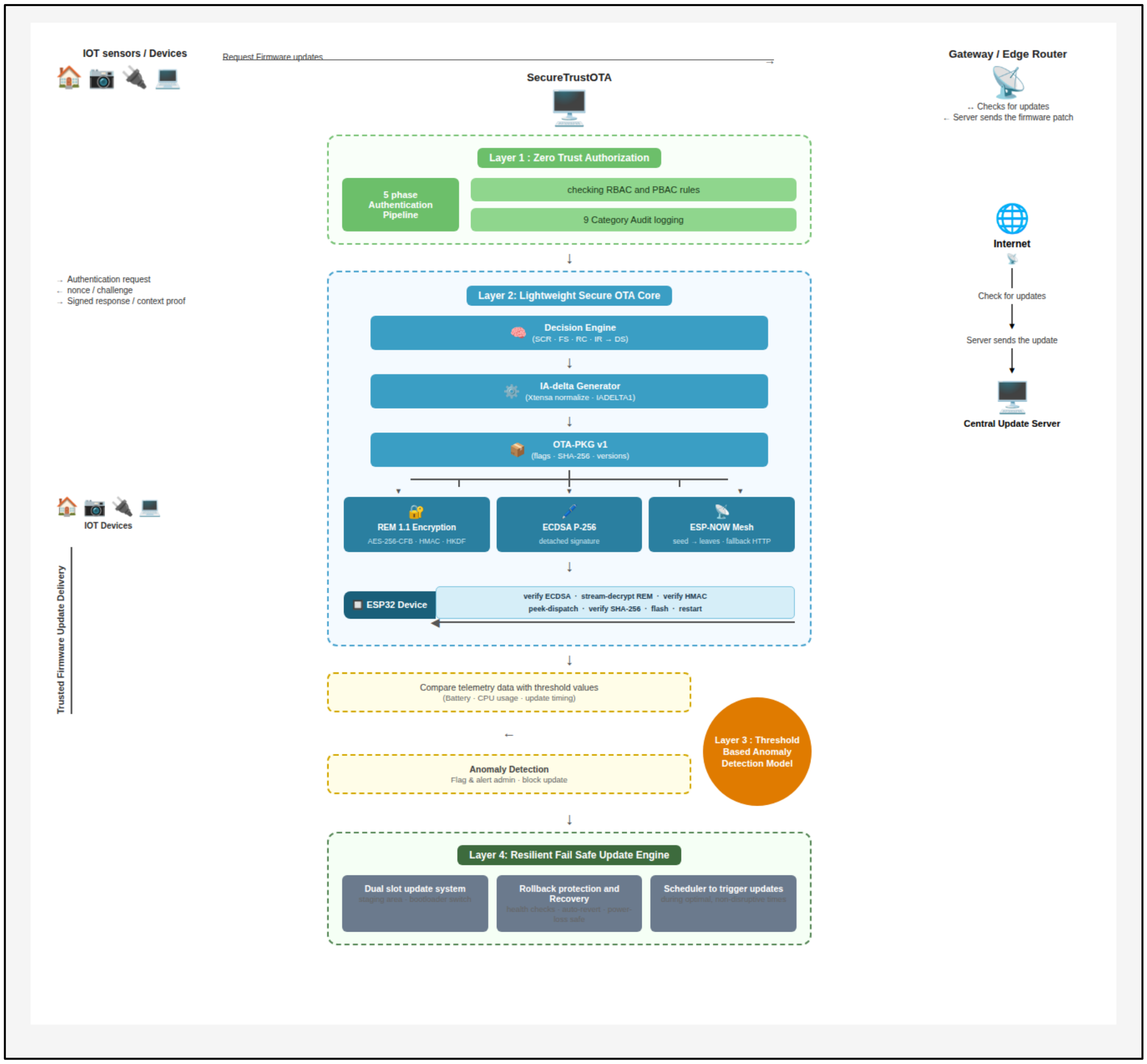

Figure 1: System Architecture

The Secure Trust OTA framework is designed as a multi-layered architecture that integrates IoT devices, secure communication channels, centralized processing, anomaly detection, and update management into a unified system. At the device level, IoT nodes are equipped with lightweight telemetry agents responsible for continuously monitoring system parameters such as CPU usage, memory utilization, battery level, storage availability, network latency, and error rates. These parameters provide a comprehensive representation of the device's operational state and are essential for anomaly detection.

The communication layer serves as a secure bridge between IoT devices and the OTA server. Protocols such as MQTT over TLS or HTTPS are used to ensure encrypted data transmission, while mutual authentication mechanisms, including certificate-based authentication, are implemented to verify device identity. This prevents unauthorized devices from interacting with the system and ensures the integrity and authenticity of transmitted data.

The OTA server functions as the central control unit, responsible for managing firmware updates, processing telemetry data, and executing anomaly detection algorithms. It maintains a centralized database for storing telemetry logs, threshold configurations, and system events. The server also hosts the anomaly detection engine, which evaluates incoming telemetry data in real time and determines whether devices are operating within acceptable limits.

A Zero Trust security layer is integrated into the architecture to enforce continuous validation of device trustworthiness. Unlike traditional systems that rely on initial authentication, this layer ensures that every update request is evaluated based on the current state of the device. This approach eliminates implicit trust and enhances overall system security.

The architecture also includes a fail-safe update mechanism that provides rollback capabilities in case of update failures. This mechanism utilizes dual-slot firmware storage, allowing devices to revert to a previous stable version if the new update fails. Additionally, update scheduling and load balancing techniques are implemented to minimize network congestion and ensure efficient distribution of updates across large device fleets.

Threshold-Based Anomaly Detection

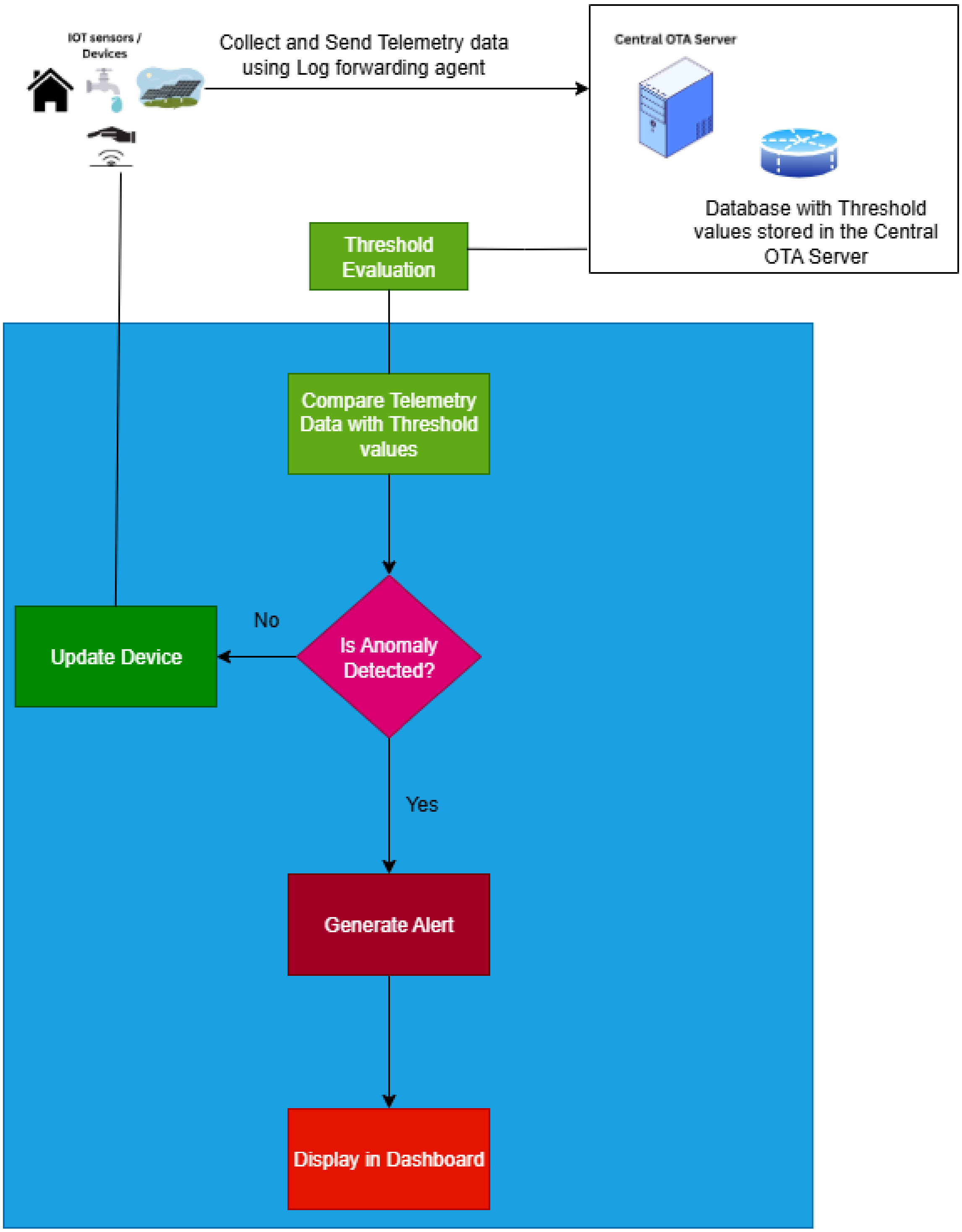

Figure 2: Model Architecture

Conceptual Framework

The threshold-based anomaly detection engine is the core component of the proposed system, designed to evaluate the operational health of IoT devices using a rule-based approach. Unlike machine learning models that require extensive training and computational resources, this approach relies on predefined thresholds to identify deviations from normal behavior. The simplicity and efficiency of this method make it particularly suitable for real-time deployment in resource-constrained environments.

The conceptual framework involves continuously monitoring device telemetry and comparing it against predefined threshold values. If the observed values exceed or fall below acceptable limits, the system classifies the behavior as anomalous. This classification is then used to determine whether the device is eligible for receiving OTA updates.

Telemetry Data Acquisition and Processing

Telemetry data acquisition is a critical component of the anomaly detection process. Each IoT device is equipped with a lightweight monitoring agent that collects system metrics at regular intervals. These metrics include CPU utilization, memory usage, battery level, storage capacity, network latency, packet loss, and error logs. The collected data is preprocessed at the device level to remove noise and ensure consistency before being transmitted to the OTA server.

Data transmission is performed using secure communication protocols to prevent interception or tampering. Upon وصول data at the server, it is stored in a time-series database, enabling efficient retrieval and analysis. The system supports high-frequency data collection, allowing real-time monitoring of device behavior and rapid detection of anomalies.

Threshold Definition and Calibration

Threshold values are defined based on device specifications, historical performance data, and operational requirements. These values represent the acceptable range of operation for each parameter. For example, CPU usage may be limited to a maximum of 85%, while battery levels may have a minimum threshold of 20% to ensure sufficient power for update operations.

Threshold calibration is performed using statistical analysis of historical telemetry data. Techniques such as mean and standard deviation calculations, percentile-based limits, and moving averages are used to determine appropriate threshold values. This ensures that thresholds are realistic and aligned with actual device behavior.

Adaptive Thresholding Mechanism

To address the limitations of static thresholds, the system incorporates adaptive thresholding techniques that dynamically adjust threshold values based on changing conditions. This is achieved باستخدام rolling window analysis, where recent telemetry data is analyzed to update threshold values periodically. Adaptive thresholding reduces false positives and improves detection accuracy by accounting for natural variations in device behavior.

Anomaly Detection Logic and Decision Engine

The anomaly detection logic is implemented as a rule-based decision engine that evaluates incoming telemetry data against predefined thresholds. For each parameter, the system checks whether the value lies within the acceptable range. If a violation is detected, the system increments an anomaly score associated with the device.

To enhance reliability, the system employs debounce logic, which requires multiple consecutive violations before confirming an anomaly. This prevents transient fluctuations from triggering false alarms. Additionally, the system supports multi-parameter correlation analysis

OTA Update Gating Integration

The anomaly detection engine is tightly integrated with the OTA update mechanism through a gating system. Before initiating an update, the OTA server evaluates the device's anomaly status. If the device is classified as normal, the update is allowed to proceed. Otherwise, the update is blocked, and an alert is generated.

This gating mechanism ensures that updates are only applied to devices operating in stable conditions, reducing the risk of update failures and system instability. It also aligns with Zero Trust principles by requiring continuous validation before granting access to critical operations.

Alert Generation and Monitoring System

When anomalies are detected, the system generates alerts that are sent to a centralized monitoring dashboard. These alerts include detailed information such as device ID, affected parameters, threshold violations, and severity levels. The dashboard provides real-time visibility into system status, enabling administrators to quickly identify and resolve issues.

Performance Optimization and Resource Efficiency

The threshold-based approach is inherently efficient, requiring minimal computational resources compared to machine learning models. The system is optimized for real-time processing, with detection latency kept below 100 milliseconds. Resource utilization is minimized to ensure scalability, allowing the system to handle large numbers of devices without performance degradation.

Limitations and Future Enhancements

While the threshold-based approach is effective, it has limitations in handling highly dynamic environments where behavior patterns may change rapidly. Future enhancements may include hybrid approaches that combine thresholding with lightweight machine learning techniques, as well as edge-based detection mechanisms to further improve scalability and responsiveness.

Commercialization Aspect of the Product

The Secure Trust OTA framework presents significant commercialization opportunities due to the increasing demand for secure IoT solutions across various industries. Organizations are increasingly adopting IoT technologies to improve efficiency and reduce costs, but concerns about security and reliability remain major barriers to adoption. The proposed system addresses these concerns by providing a robust and lightweight solution for secure firmware updates.

The system can be offered as a cloud-based service, enabling organizations to manage and monitor their IoT devices from a centralized platform. Key features include real-time anomaly detection, secure OTA updates, centralized dashboard monitoring, and alert management. This makes the solution attractive for industries such as healthcare, where device reliability is critical, and industrial automation, where downtime can result in significant financial losses.

From a business perspective, the system can be monetized through subscription-based models, licensing agreements, or integration with existing IoT platforms. The lightweight design also makes it suitable for consumer IoT devices, allowing manufacturers to enhance product reliability without increasing hardware costs.

Furthermore, the integration of security, reliability, and scalability positions the Secure Trust OTA framework as a competitive solution in the growing IoT security market. As organizations continue to prioritize cybersecurity, the demand for solutions that ensure safe and reliable device updates is expected to increase, creating opportunities for commercialization and further development.

Testing and Implementation

Implementation Overview

The implementation of the proposed Secure Trust OTA framework focuses on developing a functional prototype that integrates telemetry collection, threshold-based anomaly detection, secure OTA update management, and alert generation into a unified system. The implementation is designed to reflect real-world IoT environments while maintaining simplicity and scalability, ensuring that the solution remains practical for deployment in resource-constrained devices. A modular development approach is adopted, where each component of the system is implemented independently and later integrated to form a cohesive architecture.

The system is developed using widely adopted technologies to ensure compatibility and ease of replication. The backend OTA server and anomaly detection engine are implemented using Python due to its flexibility, extensive library support, and ease of integration with networking and data processing modules. Communication between IoT devices and the server is established using the MQTT protocol, which is lightweight and suitable for low-bandwidth environments. A time-series database is used to store telemetry data efficiently, enabling real-time analysis and historical trend evaluation.

The implementation environment simulates multiple IoT devices using virtual instances, allowing controlled testing of different operational scenarios. Each simulated device runs a telemetry agent responsible for collecting system metrics and transmitting them to the OTA server at predefined intervals. The OTA server processes incoming data, applies anomaly detection rules, and manages firmware update decisions based on the results.

System Components Implementation

The system is composed of several key modules, each responsible for specific functionalities within the framework. The telemetry collection module is implemented as a lightweight agent embedded within each IoT device. This agent periodically gathers system metrics such as CPU utilization, memory usage, battery level, and network performance. The collected data is formatted into structured messages and transmitted securely to the OTA server.

The communication module is implemented using MQTT over TLS to ensure secure data transmission. TLS encryption provides confidentiality and integrity, while client authentication ensures that only authorized devices can communicate with the server. The MQTT broker acts as an intermediary, facilitating efficient message exchange between devices and the OTA server.

The anomaly detection module is implemented as a rule-based engine within the OTA server. This module receives telemetry data, compares it against predefined threshold values, and determines whether the device is operating within acceptable limits. The implementation includes support for multi-parameter evaluation, debounce logic, and adaptive thresholding, ensuring accurate and reliable detection of anomalies.

The OTA update management module handles firmware distribution and update execution. It maintains a repository of firmware versions and manages update requests from devices. Before initiating an update, the module checks the anomaly status of the device. If the device is classified as normal, the update is allowed; otherwise, it is blocked. This module also includes rollback functionality, enabling devices to revert to a previous stable version in case of update failure.

The alert generation module is responsible for notifying administrators of detected anomalies. Alerts are generated in real time and displayed on a monitoring dashboard, which provides a comprehensive view of system status, device health, and update activities. Logs are also maintained for auditing and analysis purposes.

Testing Strategy

A comprehensive testing strategy is adopted to evaluate the functionality, reliability, and performance of the proposed system. The testing process is divided into several phases, including unit testing, integration testing, system testing, and performance testing. Each phase is designed to validate specific aspects of the system and ensure that all components operate as expected.

Unit testing is conducted on individual modules to verify their functionality in isolation. For example, the telemetry agent is tested to ensure accurate data collection and transmission, while the anomaly detection engine is tested to validate its ability to correctly identify threshold violations. Integration testing is then performed to ensure that different modules interact correctly and data flows seamlessly between components.

System testing evaluates the overall behavior of the framework under realistic conditions. This includes testing the complete OTA update process, from telemetry collection to anomaly detection and update execution. Various scenarios are simulated to assess the system's ability to handle normal and abnormal conditions.

Performance testing focuses on evaluating the system's scalability and efficiency. Metrics such as detection latency, system throughput, and resource utilization are measured to determine whether the system can handle large-scale deployments. Stress testing is also conducted by increasing the number of simulated devices to evaluate system stability under high load conditions.

Test Scenarios and Execution

Multiple test scenarios are designed to evaluate the effectiveness of the anomaly detection engine and the OTA update gating mechanism. These scenarios include both normal and abnormal operating conditions to ensure comprehensive coverage.

In normal operation scenarios, all telemetry parameters remain within predefined thresholds. The system is expected to classify devices as stable and allow OTA updates to proceed without interruption. These tests confirm that the system does not interfere with normal operations and maintains efficiency.

In abnormal scenarios, specific parameters are intentionally manipulated to simulate anomalies. For example, CPU usage is artificially increased beyond the threshold to simulate resource exhaustion, while battery levels are reduced to simulate power constraints. Network latency is also increased to simulate connectivity issues. In each case, the system is expected to detect the anomaly and block OTA updates.

Additional scenarios include multi-parameter anomalies, where multiple metrics exceed thresholds simultaneously. These tests evaluate the system's ability to detect complex conditions and ensure that the anomaly detection logic remains robust. Edge cases, such as intermittent threshold violations and missing telemetry data, are also tested to evaluate system resilience.

Evaluation Metrics

The effectiveness of the system is evaluated using several key metrics. Detection accuracy measures the system's ability to correctly identify anomalous and normal conditions. A high accuracy indicates that the threshold-based approach is effective in distinguishing between stable and unstable device states.

The false positive rate measures the frequency of normal conditions being incorrectly classified as anomalies. Minimizing false positives is important to avoid unnecessary update blocks and maintain system efficiency. Detection latency measures the time taken by the system to identify anomalies after receiving telemetry data. Low latency is essential for real-time operation.

Resource utilization is also evaluated to ensure that the system remains lightweight and scalable. Metrics such as CPU usage, memory consumption, and network bandwidth are measured to assess system efficiency. These metrics confirm that the proposed approach is suitable for large-scale IoT deployments.

Result and Discussion

Results

The evaluation of the proposed Secure Trust OTA framework was conducted through a series of controlled experiments designed to assess its effectiveness in detecting anomalies and ensuring safe firmware updates in IoT environments. The results obtained from these experiments demonstrate that the integration of threshold-based anomaly detection into the OTA update process significantly improves system reliability and operational safety.

Under normal operating conditions, where all telemetry parameters remained within predefined threshold limits, the system consistently classified devices as stable and allowed firmware updates to proceed without interruption. The update process was executed efficiently, with no noticeable delays or performance degradation introduced by the anomaly detection engine. This confirms that the proposed system does not interfere with standard operations and maintains seamless OTA functionality when devices are operating under expected conditions.

In contrast, during abnormal scenarios, the system successfully identified deviations in telemetry data and responded appropriately by blocking OTA updates. For instance, when CPU utilization exceeded the defined threshold due to simulated workload stress, the system detected the anomaly within milliseconds and prevented the update from being initiated. Similarly, when memory usage approached critical levels or when battery levels dropped below the minimum threshold required for safe update execution, the system flagged these conditions as anomalies and restricted update access. These results highlight the system's ability to proactively prevent updates that could lead to failures or system instability.

The system also demonstrated robustness in handling complex scenarios involving multiple simultaneous anomalies. In cases where multiple parameters, such as high CPU usage and increased network latency, were triggered simultaneously, the anomaly detection engine effectively identified the combined abnormal behavior and enforced update restrictions. This indicates that the system is capable of handling real-world conditions where anomalies may not occur in isolation.

Furthermore, the system maintained consistent performance across multiple test iterations, ensuring reliability and repeatability of results. The detection latency was observed to be minimal, typically within the range of milliseconds, enabling real-time decision-making. Resource utilization remained low, confirming that the threshold-based approach is efficient and suitable for large-scale IoT deployments.

Overall, the results demonstrate that the proposed framework successfully achieves its primary objective of integrating anomaly detection into the OTA update process, thereby enhancing system reliability, preventing unsafe updates, and ensuring stable device operation.

Research Findings

The findings of this research provide valuable insights into the effectiveness and practicality of integrating threshold-based anomaly detection into OTA update mechanisms for IoT systems. One of the key findings is that lightweight, rule-based detection methods can achieve high levels of accuracy in identifying abnormal device behavior without the need for complex machine learning models. This is particularly important in IoT environments, where computational resources are limited and efficiency is critical.

Another significant finding is that proactive validation of device health before initiating firmware updates greatly reduces the risk of update failures and system instability. By introducing a gating mechanism that blocks updates on anomalous devices, the system ensures that updates are only applied under safe conditions. This approach shifts the paradigm from reactive recovery, where issues are addressed after they occur, to proactive prevention, where potential failures are avoided altogether.

The research also highlights the importance of selecting appropriate threshold values for anomaly detection. It was observed that poorly defined thresholds could lead to false positives or missed anomalies. However, the use of adaptive thresholding techniques significantly improved detection accuracy by accounting for natural variations in device behavior. This finding underscores the need for dynamic threshold management in real-world deployments.

Additionally, the study confirms that integrating anomaly detection into OTA workflows aligns well with Zero Trust security principles. By continuously validating device trustworthiness before performing critical operations, the system eliminates implicit trust and enhances overall security. This integration not only improves reliability but also strengthens the system's defense against potential cyber threats.

Another important finding is the scalability of the proposed approach. The system demonstrated the ability to handle multiple devices simultaneously without significant performance degradation. This indicates that the framework can be effectively deployed in large-scale IoT environments, making it a practical solution for industry applications.

Discussion

The results and findings of this study indicate that the proposed Secure Trust OTA framework offers a significant improvement over traditional OTA update systems. By incorporating a threshold-based anomaly detection engine into the update process, the system addresses a critical limitation in existing frameworks, namely the lack of device state validation before updates. This enhancement not only improves reliability but also contributes to overall system security.

One of the major strengths of the proposed approach is its simplicity and efficiency. Unlike machine learning-based methods, which require extensive training and computational resources, the threshold-based approach relies on straightforward comparisons that can be executed in real time with minimal overhead. This makes the system highly suitable for resource-constrained IoT environments, where efficiency is a key requirement. The ability to achieve reliable anomaly detection without complex models demonstrates the practicality of this approach for real-world applications.

The integration of anomaly detection into the OTA workflow represents a shift towards proactive system management. Traditional systems often rely on rollback mechanisms to recover from failed updates, which can be costly and time-consuming. In contrast, the proposed system prevents such failures by ensuring that updates are only applied when devices are in a stable state. This proactive approach reduces downtime, improves system reliability, and enhances user trust in IoT devices.

However, the system is not without limitations. One of the primary challenges is the reliance on predefined threshold values, which may not always accurately reflect dynamic changes in device behavior. While adaptive thresholding improves flexibility, there is still a possibility of false positives or false negatives in highly variable environments. Future work could explore hybrid approaches that combine threshold-based methods with lightweight machine learning techniques to enhance detection accuracy.

Another limitation is the centralized nature of the anomaly detection process, which may introduce latency in extremely large-scale deployments. Although the current system demonstrates low latency, further optimization may be required for environments with millions of devices. Edge-based anomaly detection could be considered as a potential solution to reduce latency and improve scalability.

Despite these limitations, the proposed framework provides a strong foundation for improving the security and reliability of OTA update mechanisms. The results demonstrate that integrating anomaly detection into the update process is both feasible and effective, offering a practical solution to a critical problem in IoT systems.

In conclusion, the discussion highlights that the Secure Trust OTA framework successfully bridges the gap between anomaly detection and OTA update mechanisms. By ensuring that updates are applied only under safe conditions, the system enhances reliability, reduces risks, and aligns with modern cybersecurity practices. This makes it a valuable contribution to the field of IoT security and a promising direction for future research and development.

Summary of Contribution

This research makes several significant contributions to the field of IoT security and firmware update management by addressing a critical limitation in existing Over-the-Air (OTA) update mechanisms. The primary contribution of this study is the design and development of a Secure Trust OTA framework that integrates a threshold-based anomaly detection engine into the firmware update process. This integration introduces a proactive validation mechanism that ensures updates are applied only when IoT devices are operating under stable and trusted conditions.

One of the key contributions of this work is the introduction of a lightweight anomaly detection approach tailored specifically for resource-constrained IoT environments. Unlike traditional methods that rely on computationally intensive machine learning algorithms, the proposed threshold-based method provides an efficient and practical alternative for real-time anomaly detection. By leveraging predefined and adaptive threshold values, the system is able to accurately identify abnormal device behavior while maintaining low computational overhead, making it suitable for large-scale deployments.

Another important contribution is the development of an OTA update gating mechanism that directly links anomaly detection outcomes with update decision-making. This mechanism prevents firmware updates from being applied to devices exhibiting anomalous behavior, thereby reducing the risk of update failures, system crashes, and potential security breaches.

By continuously validating the operational state of devices before granting permission for firmware updates, the system eliminates implicit trust and enforces strict verification policies. This enhances the overall security posture of the IoT ecosystem and aligns with modern cybersecurity best practices.

In addition, the study provides a comprehensive system architecture that integrates telemetry collection, secure communication, anomaly detection, and update management into a unified framework. The modular design of the system ensures scalability, flexibility, and ease of integration with existing IoT platforms. The implementation and testing of the system further demonstrate its feasibility and effectiveness in real-world scenarios, highlighting its potential for practical deployment.

Furthermore, the research offers valuable insights into threshold selection, adaptive thresholding techniques, and performance optimization for anomaly detection in IoT systems. The findings emphasize the importance of balancing detection accuracy and system efficiency, providing a foundation for future research in this area.

Overall, this study contributes a novel, practical, and scalable solution to enhance the security and reliability of OTA firmware updates in IoT environments. By addressing the gap between anomaly detection and update management, the proposed framework advances the state of the art and provides a strong basis for future innovations in IoT security.

Conclusion

The rapid expansion of Internet of Things (IoT) ecosystems has introduced significant challenges in maintaining the security, reliability, and manageability of connected devices. Among these challenges, the need for secure and efficient firmware update mechanisms has become increasingly critical, as vulnerabilities in outdated firmware can expose devices to cyber threats and operational failures. Over-the-Air (OTA) update systems have emerged as a practical solution for remotely managing firmware updates; however, existing OTA frameworks primarily focus on secure transmission and integrity verification of update packages while overlooking the operational state of the devices receiving these updates. This limitation creates a significant risk, as applying updates to devices operating under unstable or compromised conditions can result in system failures, data corruption, or security breaches.

This research addressed this critical gap by proposing a Secure Trust OTA framework that integrates a threshold-based anomaly detection mechanism into the OTA update process. The primary objective was to ensure that firmware updates are applied only when IoT devices are operating within safe and stable conditions. By incorporating real-time telemetry monitoring and anomaly detection into the update workflow, the system introduces a proactive validation layer that enhances both reliability and security.

The proposed framework was designed using a modular and scalable architecture that includes telemetry collection, secure communication, centralized processing, anomaly detection, and OTA update management. The threshold-based anomaly detection engine plays a central role in the system by evaluating key operational parameters such as CPU usage, memory utilization, battery levels, and network performance. By comparing these parameters against predefined and adaptive threshold values, the system is able to accurately identify abnormal behavior in real time. The integration of this detection mechanism with the OTA update process enables a gating system that blocks updates for devices exhibiting anomalous conditions, thereby preventing potential failures.

The implementation and testing of the system demonstrated that the proposed approach is both effective and efficient. The results showed that the system successfully detects anomalies with high accuracy while maintaining low latency and minimal resource consumption. Under normal conditions, firmware updates were executed seamlessly without any performance degradation, confirming that the anomaly detection mechanism does not interfere with standard operations. In abnormal scenarios, the system consistently identified deviations in device behavior and prevented updates, thereby reducing the risk of system instability and failure.

The findings of this research highlight several important contributions. First, the study demonstrates that lightweight, threshold-based anomaly detection methods can provide reliable real-time monitoring without the need for computationally expensive machine learning models. This makes the approach highly suitable for resource-constrained IoT environments. Second, the integration of anomaly detection into OTA workflows represents a shift from reactive to proactive system management, significantly improving reliability and reducing operational risks. Third, the application of Zero Trust principles in the update process enhances security by ensuring continuous validation of device trustworthiness.

Despite these contributions, the study also acknowledges certain limitations. The effectiveness of the threshold-based approach depends on the accuracy of predefined threshold values, which may not fully capture dynamic changes in device behavior. Although adaptive thresholding techniques were implemented to address this issue, there is still potential for improvement in highly variable environments. Additionally, the centralized nature of the system may introduce scalability challenges in extremely large deployments, suggesting the need for further optimization or the incorporation of edge-based processing.

Future work can build upon this research by exploring hybrid anomaly detection models that combine threshold-based methods with lightweight machine learning techniques to improve detection accuracy. The integration of edge computing can also be considered to reduce latency and enhance scalability. Furthermore, extending the framework to support additional security features, such as intrusion detection and automated response mechanisms, could further strengthen the system's capabilities.

In conclusion, this research presents a novel and practical solution to a critical problem in IoT systems by integrating anomaly detection into OTA update mechanisms. The proposed Secure Trust OTA framework enhances the reliability, security, and efficiency of firmware updates, providing a strong foundation for future advancements in IoT security. Its lightweight design, scalability, and proactive approach make it a valuable contribution to both academic research and real-world applications, supporting the development of more secure and resilient IoT ecosystems.

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